International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE
Mid term Exam, Spring 2022

Course Code: **CSE 3525** Course Title: **Data Communication**
Time: 1 hour and 30 minutes Full Marks: 30

(The figures in the right-hand margin indicate full marks)

1. a) Following figure (Figure 1) shows that the communication is between a [illegible] computer R with port address k. S[illegible] content of transport, network and data link layers for each hop. **CLO1** 5

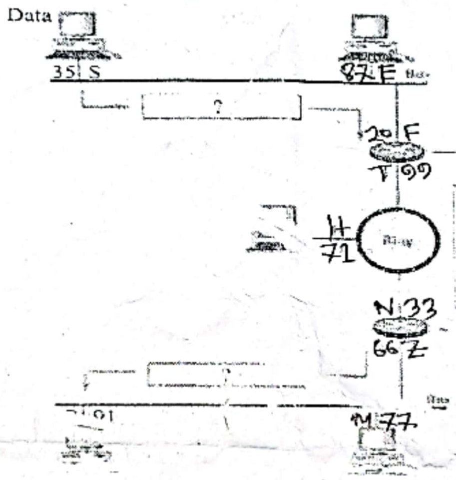


Figure 1

1 b) Describe the functions of each of the layers of ISO-OSI model. **CLO1** 5

2. a) Show the time and frequency domain plot of a voice signal and a sine wave. Also calculate and show the bandwidth for both signals. **CLO1** 5

2. b) A signal at the end of a 3 km transmission media has a power of 3mW. If the loss in the cable is 2 db, what was the power of the signal at the starting point of the media. **CLO2** 5

OR

2 b) Analyze, compare and evaluate the formula for Nyquist bit rate for a noiseless channel and the formula for Shannon capacity for a noiseless channel. **CLO2** 5

3. a) Write differences between synchronus and asynchronus transmission. Discuss signal propagation through guided and unguided media **CLO2** 5

3. b) With figure show any two method of converting digital data to digital signal. **CLO3** 5

OR

3 b) Construct digital signal with the following line coding schemes for the following bit stream: 100100000111 **CLO3** 5

i. NRZ-I
ii. Manchester
iii. Differential Manchester
iv. AMI
v. AMI with HDB3

International Islamic University Chittagong

Department of Computer Science and Engineering

Mid Examination, Spring 2022

Course Code: CSE 3635 | **Course Title: Artificial Intelligence**
Total Marks: 30 | **Time: 1.5 Hours**
Semester:6th

<u>**Answer the following questions**</u>

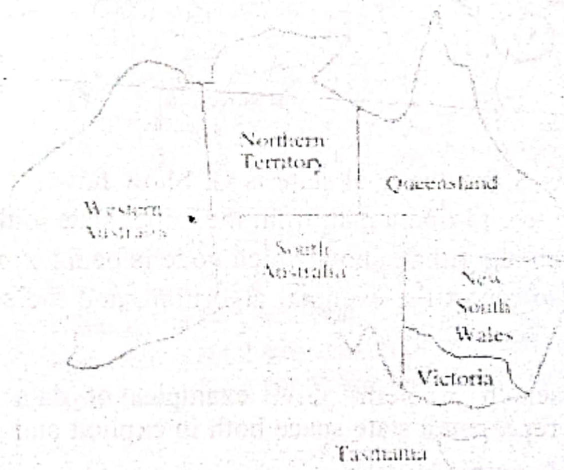
1. a) Suppose you are going to create a smart agent that picks different types of parts from a floor and put into bins. The goal is to separate the parts according to their types. Sometimes, the floor could be slippery. Write the PEAS description of this intelligent agent. 2 CO1

OR

a) Give a scientific definition of intelligence. Show how does it differ from the dictionary definition of intelligence. What methods have been used to define artificial intelligence? Which one, in your opinion, is appropriate? Why?

b) Identify the environment of the part picking agent as deterministic vs stochastic and static versus dynamic. 1 CO1

c) Consider the Australian map below. You want to color areas so that none of their neighbors have the same color. The four colors you may choose from are Red, Blue, and Green. Start from western Australia. When you are in an area, you may go to nearby regions or color the current region. Identify the variables, domains and constraints associated with this problem. 5 CO1

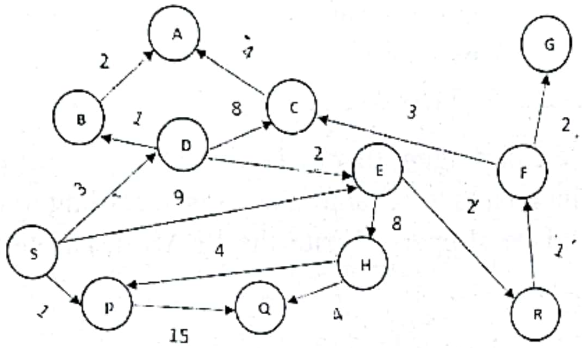


d) Differentiate the following algorithms by taking account of the factors such as completeness, optimality, space complexity and time complexity 2 CO2

i) BFS, ii) DFS, iii) DLS, iv) UCS

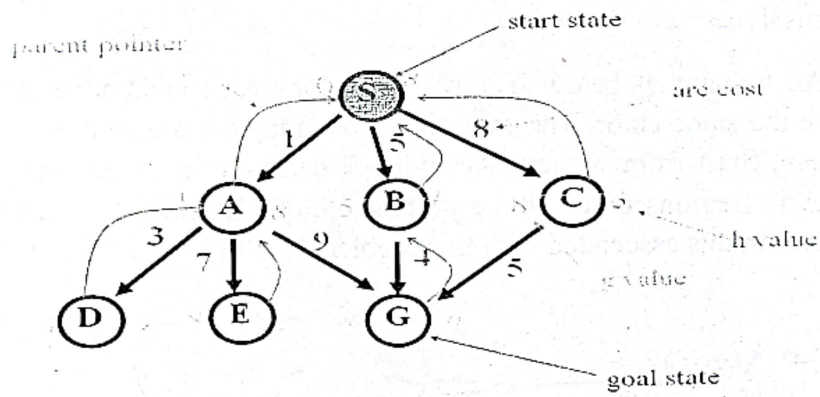
WA = R B G BG BG BG BG BG
NT = B R B
SA = R B G
Q = R R B G R
NSW = B R B G B B
V = R. R. B. G. B. R

2. a) Take a look at the state-space graph in the following figure. The start node is S. while the destination node is G. Draw the search tree for the uniform cost search method, then find out the solution paths and costs. 5



OR

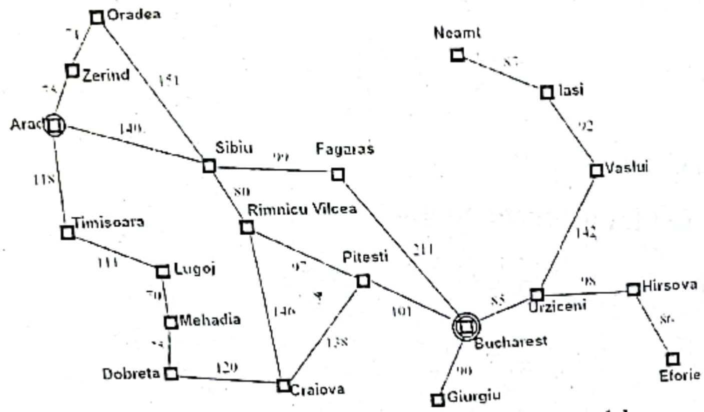
a) Suppose you have the following search space:



Assume that the initial state is S and the goal state is G. Show how A* search strategy would create a search tree to find a path from the initial state to the goal state. At each step of the search algorithm, show which node is being expanded, and the content of fringe. Also report the eventual algorithm, and the solution cost.

b) How can you formulate a search problem? Give examples of data structures that can be used to represent a state space both in explicit and implicit ways. 2 CO1

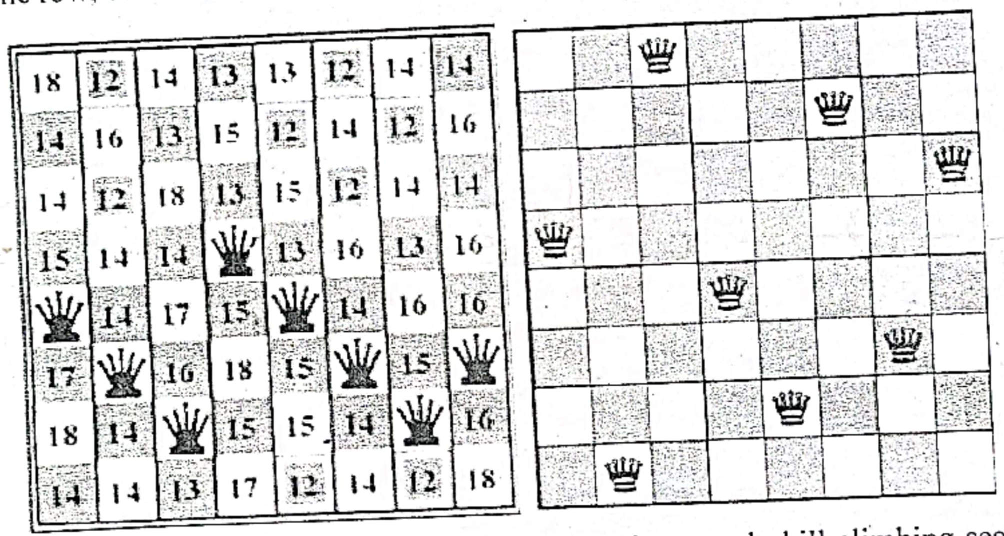
c) Consider the following graph where you have to travel from Arad to Bucharest. 3 CO2



Now formulate the problem as a search problem.

3. a) What do you mean by Knowledge Representation and mapping? What are the roles of Knowledge Representation? 2 CO3

b) Consider the 8-queen problem where you have a board with 8 column and 8 rows. Your objective is to arrange the queens so that no two queens share the same row, column, or diagonal. Consider the following start state and goal state. 3 CO3



Suppose you are trying to solve this problem using greedy hill climbing search. According to this strategy find the followings:

i. Write a suitable heuristic function.
ii. What is the heuristic value of this start space?
iii. How many total successors of the start space?
iv. How to achieve 100% success for these types of problem?

OR

b) Write down Steepest-Ascent Hill Climbing Algorithm. What are the problems that the hill climbing search process may reach? How the problems can be dealt?

c) Prove FOL is considered as the generalization of PL. 5 CO3

Translate the following sentences into FOL

i. All dogs are faithful

ii. Mammals drink milk
iii. Some birds cannot fly
iv. There are books that are heavy.

v. If it is an animal then it is an organism

vi. Each student must be registered for a degree programme.

vii. Everyone is loyal to someone.

Bismillaher Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Spring-2022

Course Code: CSE-3631 Course Title: Operating System

Time: 1 hour and 30 minutes Total marks: 30

[Answer all the questions. Figures in the right hand margin indicate full marks.]

	Question	Marks	CO	DL
1.				
a)	Do you think that Operating System acts as a *file manager*-----how?	2	CO1	C1
b)	Mention the differences between *clustered system* and *desktop system.*	2	CO1	C2
	OR			
b)	Describe *real time system* with example.	2	CO1	C2
c)	What is *virtual machine* and what are the benefits of it ?	2	CO1	C2
d)	Operating system is a very large software but your RAM is very small in size. We are at the age of multiprogramming system. Different types of applications we have to run at a time. How operating system can give the accommodation of these applications at a time to the RAM?	4	CO2	C3
2.				
a)	What are the differences between *ready state* and *waiting state* of a job and what is the main function of *dispatcher*?	2	CO1	C2
	OR			
a)	Write the advantages of *job control language.*	2	CO1	C2
b)	What do you mean by *buffering* of communication between jobs?	2	CO3	C2
c)	What is better between *deterministic modelling* and *queueing modelling* for jobs and how?	4	CO3	C2
	OR			
c)	What are the differences between *deterministic modelling* and *simulation*?	4	CO3	C2
d)	Why *waiting time* is essential parameters in CPU job scheduling?	2	CO3	C2
3.				
a)	Find the *average waiting time* and *average turnaround time* for the following jobs by Round Robin job scheduling algorithm:	5	CO4	C3

SL. of Jobs	Arrival time	Service time
P1	1	15
P2	8	8
P3	10	1
P4	15	1
P5	18	7

(Here quantum size is 4)

b) Find the *average waiting time* and *average turnaround* time for the following jobs by First Come First Served job scheduling algorithm: 5 CO4 C3

SL. of Jobs	Arrival time	Service time
P1	1	12
P2	7	7
P3	11	6
P4	12	9
P5	17	8

OR

b) Describe *processor access methods* in respect of operating system. 5 CO1 C1

END

International Islamic University Chittagong

Morality Development Program (MDP)

Mid Term Examination, Spring-2022

Faculty of Science and Engineering, Semester- 6th

Course Title: Islamization of Discipline **Course Code:** MDP-3606

Time:1:30 Hours **Total Marks-30**

Figures in the right hand margin indicate full marks

Answer any ***three*** questions

1. a) What is Black-Hole? Mention one Quranic verse in favor of Black-Hole 2

b) What is the *Hubble's constant?* What are the functions of 'Hubble Space Telescope' 2

c) Mention one Quranic Verse regarding Big-Bang and Big Crunch 2

2. a. What do you mean by Open Universe, Closed Universe and Flat Universe? 5

b. What is singularity and Big-bang of the universe and Dark matter? 5

3. Discuss the scientific analysis of the human embryonic stages in the light of the Holy Quran 10

4. a) What is the challenge of the Quran? Mention any Quranic Verse in favor of this. **2+2**

b) *"Scientists have discovered water barriers which divide between salty and sweet water in downstream areas."* Mention any Quranic Verse in favor of this **2**

c) Is the light of the Moon reflected light? How? Mention any Quranic Verse in favor of this. **2+2**

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. Engineering in CSE

Midterm Examination. Spring 2022

Course Code: ECON 3501 Course Title: Principle of Economics

Time: 1 hour 30 minutes Full Marks: 30

Answer all the questions. The figures in the right-hand margin indicate full marks.

1)	a)	"Economic problem consists in making decisions regarding the wants to be pursued and the resources to be used for the achievement in satisfying wants."- Explain the fundamental Economic problems with example. Draw a detail circular flow model of economic activity that shows the basic relationships between two vital factors of economy.	CO1	C	5
1)	b)	Discuss the micro and macro goals of economics with example.	CO1	U/E	5
2)	a)	What is Equilibrium? Explain the market equilibrium using demand and supply curve with an example? The demand function of two commodities A and B are $D_A = 20 - P_A - 2P_B$, $D_B = 16 - P_A - P_B$ and the corresponding supply functions are $S_A = -3 + P_A + P_B$, $S_B = -2 + P_B$. Where P_A and P_B denote the prices of A and B respectively. Find i. The equilibrium prices. and ii. The equilibrium quantities exchanged in the market.	CO2	An/ E	5
2)	b)	Discuss the types of elasticity of demand with example. Suppose the price of coffee rises from Tk. 6.00 per 100 grams to Tk.8 per 100 grams and as a result the consumers demand for tea increases from 50 grams to 60 grams. Then find the cross elasticity of demand of Tea for Coffee.	CO2	An/ E	5
3)	a)	Explain the types of returns to-scale with example.	CO3	Un/ E	5
3)	b)	Describe indifference curve and its properties with example? How we can draw indifference curve?	CO3	E	5
		OR			
3)		Write short notes: a) Euler's Theorem b) Total product, average product and marginal product c) Production Possibility Curve d) Positive & Normative Economics e) Marginal Utility	CO3	Un/ E	2 x 5

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Mid Term Exam, Spring 2022

Course Code: **CSE-4805** Course Title: **Social, Professional, Ethical Issues in Computing**

Time: 1 hour and 30 minutes Full Marks: 30

(i) Answer all three (3) questions. The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs) of the Questions	
CO1	Remember, identify, and apply different ethical philosophies, frameworks, and methodologies. The goal is for students to be able to address ethical dilemmas with reasoned arguments, grounded in a combination of these ethical theories.
CO2	Understand and identify and interpret the codes of professional conduct relating to the disciplines of computer science and software engineering such as ACM Code of Ethics. We will also learn how we can use these in our daily practice.
CO3	Apply the concepts and principles of moral thinking to problems relating to computing and digital technologies.
CO4	Analyze the local and global impact of computing on individuals, organizations, and society. Students will be able to discuss key concepts in a digital society including issues of copyright, privacy, personal freedom, computer crimes and new legal issues as well as advances in medicine, telecommunications and education.

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1.	a)	What is computing? Discuss some professional issues in computing.	CLO1	R	5
	b)	Write down the differences between ethics and morality. Analyze social impact of computers to today's world.	CLO2	U	5
2.	a)	What does the term personal information mean? Discuss how CCTV and other electronic devices hamper our privacy? What are the remedies?	CLO4	An	5
	b)	A very large social network company analyzes all data it gathers through its service on its members' activities to develop statistical information for marketers and to plan new services. The information is very valuable. Should the company pay its members for its use of their information?	CLO4	An	5
		OR (for 2b only, 2a must answer)			
	b)	Describe two methods a business or agency can use to reduce the risk of unauthorized release of personal information by employees.	CLO4	An	5

3. a)	What is legality? Suppose, Mr. X is walking in the street. And he found some money dropped in the street, and keep it. Determine his action of behavior in case of legality and ethicality. Justify your answer based on that issue.	CLO4	An	5
b)	Suppose, Mr. Y got a job opportunity and joined in the job. But, the job isn't permanent based on appointment. Then he got an offer from other corporate job with higher salary. So, Mr. Y left the current job and joined in the corporate job. i. Determine the act of Mr.Y based on teleological theories. Briefly explain it. ii. Determine his action based on legality and ethicality. iii. What are the criticisms for this course of action by Mr. Y?	CLO3	App	5
	OR (for 3b only, 3a must answer)			
b)	Briefly discuss the key points of ICT act of Bangladesh. Do you have any social responsibility to spread ICT act among general peoples? How do you perform those responsibilities?	CLO3	App	5

International Islamic University Chittagong
Center for General Education (CGED)
Midterm Examination- Spring-2022
Course Title: Life and Teachings of Prophet (SAAS)
Course Code: URED-3604
(URED-3201 for LLB)

Time 1:30 Hours **Full Marks: 30**

[Answer any three of the following questions and all questions are of equal value]

Q.1.

What do you understand by *Sirah*? Discuss the benefit of studying the *Sirah* of the Prophet (SAAS) in the light of the Qur'an and *Sunnah.*

Q.2.

Discuss the socio-political and religious condition of *Arabiya* before the advent of Prophet Muhammad (SAAS).

Q.3.

What obstacles did the Quraiysh put in the way of spreading the call of the Prophet Muhammad (SAAS)? Explain.

Q.4.

Talk about the migration of Muslims to Abyssinia and then analyze the sermon delivered by Ja'far bin Abi Talib before Negus, the king of Abyssinia.

Q.5.

Write short notes on the following topics: (Any two)

a. Boycott and confinement

b. *Isra'* and *Mi'raj*

c. Journey to *Ta'if*
